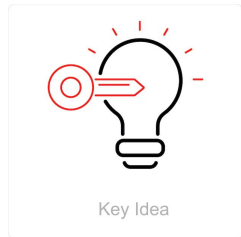




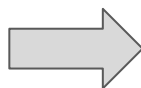
Introducción a R

Evelin González F.
evefeliu@gmail.com



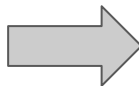
Conceptos clave

¿Qué es R?



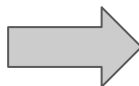
Lenguaje para analizar datos

¿Cómo guarda datos R?



Variables

¿Cómo organiza datos R?



Data frame (tabla)

	apples	oranges
0	3	0
1	2	3
2	0	7
3	1	2

¿Cómo generamos análisis reproducible?



R Markdown

Introducción a R

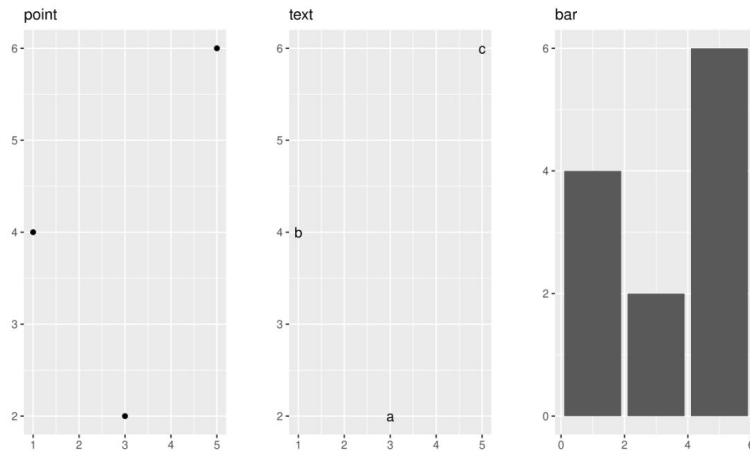


- ❖ Lenguaje para analizar datos
- ❖ Gráficos y estadística
- ❖ Reproducible
- ❖ Open source

```
df <- data.frame(  
  x = c(3, 1, 5),  
  y = c(2, 4, 6),  
  label = c("a", "b", "c")  
)  
p <- ggplot(df, aes(x, y, label = label)) +  
  labs(x = NULL, y = NULL) + # Hide axis label  
  theme(plot.title = element_text(size = 12)) # Shrink plot title  
p + geom_point() + ggtitle("point")  
p + geom_text() + ggtitle("text")  
p + geom_bar(stat = "identity") + ggtitle("bar")
```

¿Por qué usar R en bioinformática?

- Análisis de datos genómicos.
- Gráficos de calidad publicación
- Paquetes especializados (Bioconductor) Ej: **ggplot2**



Rstudio: nuestro espacio de trabajo

¿Dónde escribimos código?

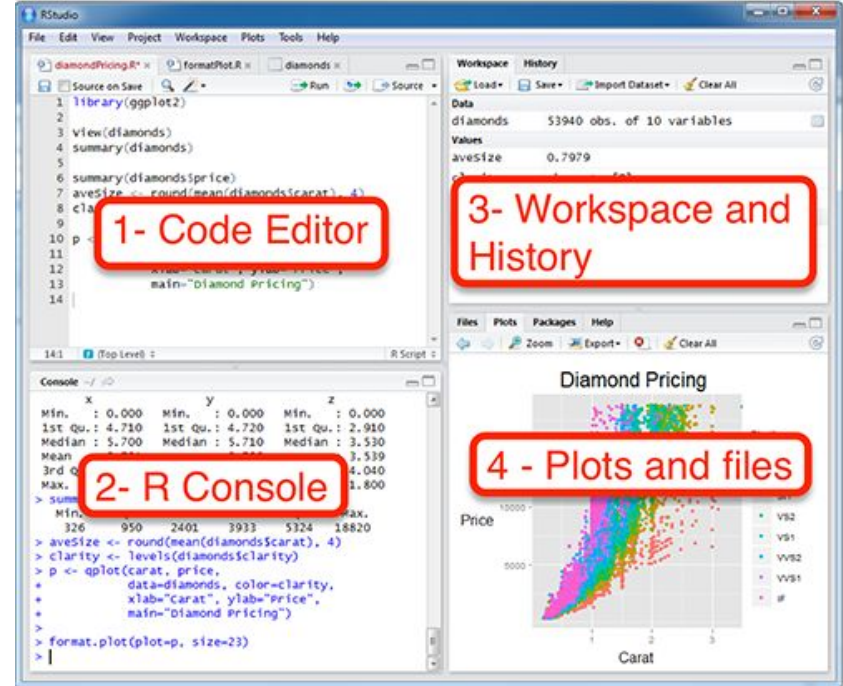
¿Dónde aparecen los objetos?

¿Dónde vemos gráficos?



Lin descarga

<https://posit.co/download/rstudio-desktop/>



Rstudio

Source Pane

Edit and run scripts (e.g. Rmarkdown templates), and view datasets

Tip: Start new script

Tip: Run script

Environment Pane

Overview of objects (datasets, parameters, lists, etc.) you have imported or created.

Tip: Zoom and export plots

R Console Pane

R commands run are shown here, and non-graphic output and errors are displayed

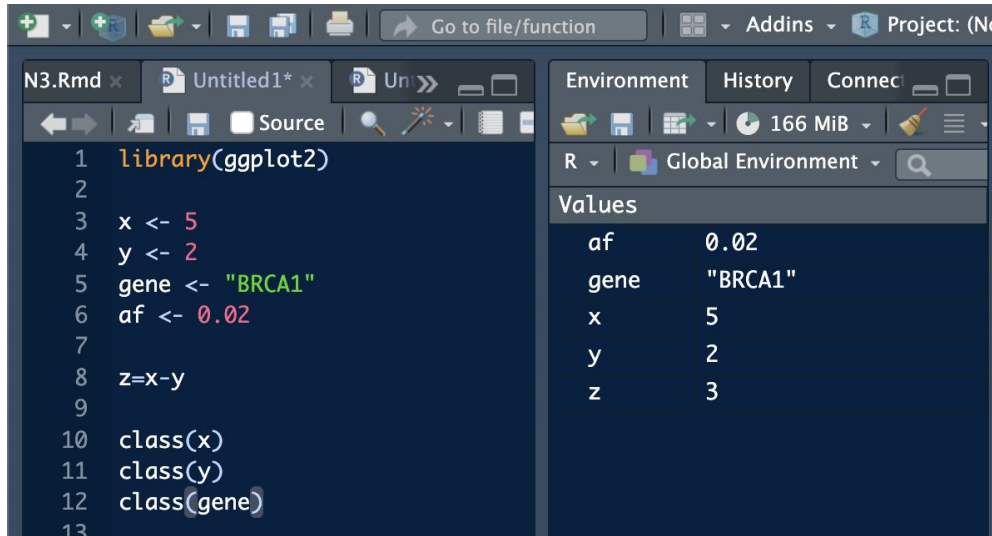
Plots, Packages, and Help Pane

Commonly used to view graphics, install packages, and view help

The screenshot displays the RStudio IDE interface. The top-left pane (Source Pane) shows an R Markdown file titled "cholera outbreak report" with a template for an automated outbreak situation report. The top-right pane (Environment Pane) lists objects in the Global Environment, including "linelist_cleaned" (300 obs. of 54 variables), "linelist_dict" (183 obs. of 11 variables), "linelist_raw" (300 obs. of 46 variables), "population_da_" (5 obs. of 3 variables), and "population_da_" (4 obs. of 3 variables). The bottom-left pane (R Console Pane) shows the execution of the `plot_age_pyramid` function, which generates an age pyramid plot. The bottom-right pane (Plots, Packages, and Help Pane) displays the resulting age pyramid plot, showing the distribution of cases by age group (0-4 to 45+) and sex (Male, Female, hown/unspec). The plot includes a legend for Confirmed (green), Probable (purple), and Suspected (orange) cases. The x-axis represents the number of cases (n), and the y-axis represents the age group (years).

Variables en R

Una variable es un nombre que guarda un valor en memoria.



The screenshot shows the RStudio interface. The Source pane on the left contains the following R code:

```
1 library(ggplot2)
2
3 x <- 5
4 y <- 2
5 gene <- "BRCA1"
6 af <- 0.02
7
8 z=x-y
9
10 class(x)
11 class(y)
12 class(gene)
13
```

The Environment pane on the right shows the current environment (Global Environment) with the following variables and values:

Variable	Value
af	0.02
gene	"BRCA1"
x	5
y	2
z	3

`class(x)` $\Rightarrow$ "numeric"

`class(gene)` $\Rightarrow$ "character"

```
v <- c(1,2,3)
```

Vectores en R

Un vector es una colección de valores del mismo tipo.

- Se crean con `c()`
- Todos los elementos deben ser del mismo tipo
- Permiten operaciones matemáticas

The screenshot shows the RStudio interface. The source editor on the left contains the following R code:

```
1  
2 v <- c(1, 2, 3, 4)  
3 genes <- c("BRCA1", "TP53", "BRCA2")  
4  
5 mean(v)  
6 length(v)
```

The Environment pane on the right shows the current state of the workspace:

Values	
genes	chr [1:3] "BRCA1" "TP53" "BRCA2"
v	num [1:4] 1 2 3 4

```
> mean(v)  
[1] 2.5
```

```
> length(v)  
[1] 4
```

Dataframe en R

Una tabla de datos donde las filas representan muestras u observaciones y las columnas representan variables.

```
1 df <- data.frame(  
2   sample=c("S1", "S2", "S3"),  
3   gene=c("BRCA1", "TP53", "BRCA2"),  
4   AF=c(0.1, 0.5, 0.02)  
5 )
```



	sample	gene	AF
1	S1	BRCA1	0.10
2	S2	TP53	0.50
3	S3	BRCA2	0.02

¿Cómo accedemos a los datos ?

```
> df$gene  
[1] "BRCA1" "TP53" "BRCA2"  
> df$AF  
[1] 0.10 0.50 0.02  
> df$sample  
[1] "S1" "S2" "S3"  
> df[1,2]  
[1] "BRCA1"  
> df$AF>0.1  
[1] FALSE TRUE FALSE  
> df[1,]  
  sample gene AF  
1    S1 BRCA1 0.1  
> df[,1]  
[1] "S1" "S2" "S3"
```

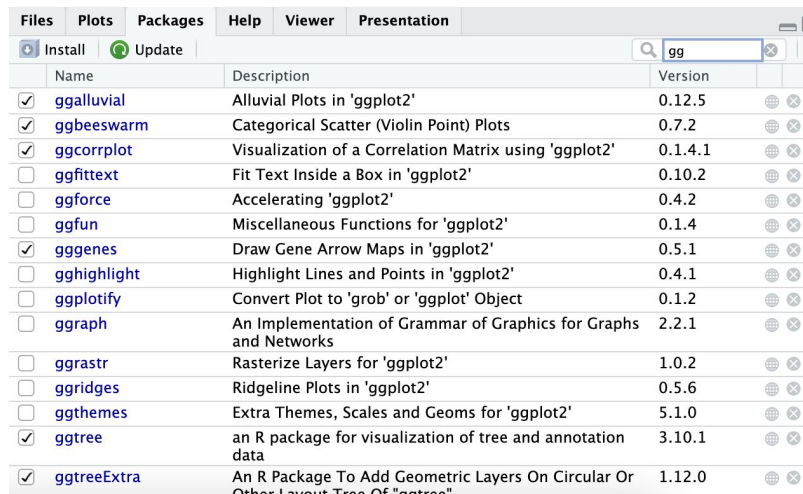
filas = muestras

columnas = variables

Paquetes en R

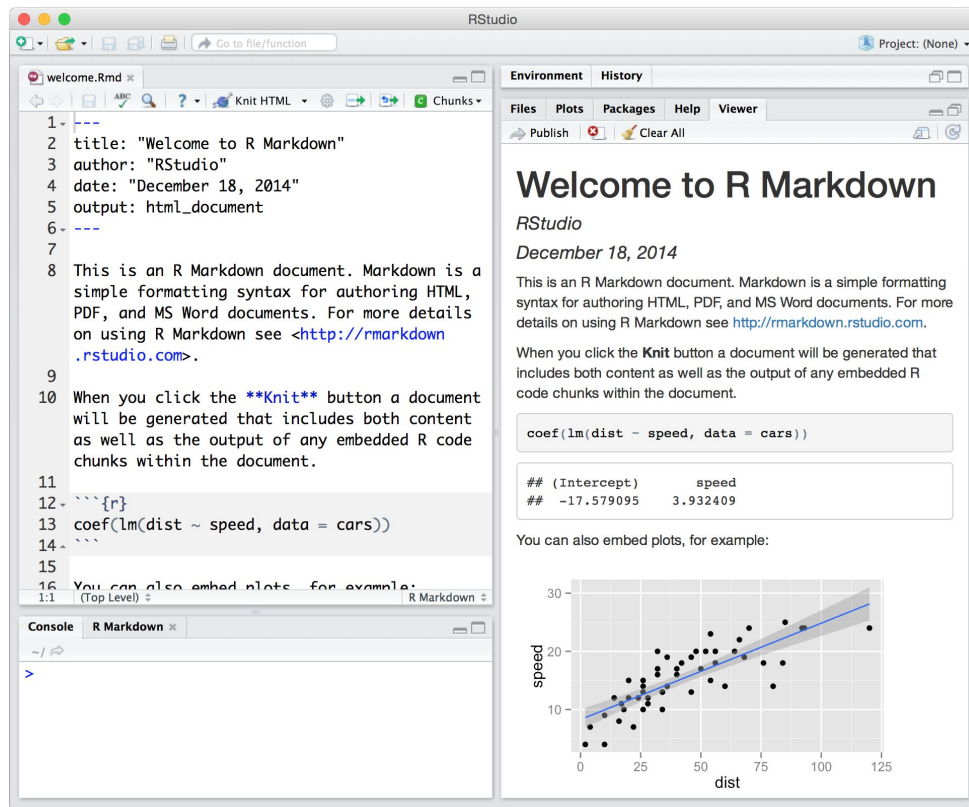
- Instalar (CRAN, devtools)
`install.packages("tydiverse")`
- Import: `library(tydiverse)`
- Recurrentes: `dplyr`, `ggplot2`, `maftools`.

```
suppressMessages(library(ape)) #tree basics
suppressMessages(library(treeio)) #join trees
suppressMessages(library(ggtree)) #tree plotting
suppressMessages(library(ggplot2)) #plots
suppressMessages(library(ggtreeExtra)) #geom_fruit
suppressMessages(library(ggnewscale)) #scales in ggplot2
suppressMessages(library(ggstar)) #geom_star for ggplot2
suppressMessages(library(tydiverse)) #strings, dataframes processing
suppressMessages(library(magrittr)) #pipes %>%
suppressMessages(library(tidytree)) #trees as dataframes
suppressMessages(library(RColorBrewer)) #color palettes
suppressWarnings(library(colorspace)) #colors manipulation
suppressWarnings(library(patchwork)) #side-by-side plotting
suppressWarnings(library(openxlsx)) #read excel
```



Files	Plots	Packages	Help	Viewer	Presentation
		Install	Update	gg	
	Name	Description	Version		
☒	ggalluvial	Alluvial Plots in 'ggplot2'	0.12.5		
☒	ggbeeswarm	Categorical Scatter (Violin Point) Plots	0.7.2		
☒	ggcorrplot	Visualization of a Correlation Matrix using 'ggplot2'	0.1.4.1		
☐	ggfittext	Fit Text Inside a Box in 'ggplot2'	0.10.2		
☐	ggforce	Accelerating 'ggplot2'	0.4.2		
☐	ggfun	Miscellaneous Functions for 'ggplot2'	0.1.4		
☒	gggenes	Draw Gene Arrow Maps in 'ggplot2'	0.5.1		
☐	gghighlight	Highlight Lines and Points in 'ggplot2'	0.4.1		
☐	ggplotify	Convert Plot to 'grob' or 'ggplot' Object	0.1.2		
☐	gggraph	An Implementation of Grammar of Graphics for Graphs and Networks	2.2.1		
☐	gggrastr	Rasterize Layers for 'ggplot2'	1.0.2		
☐	gggridges	Ridgeline Plots in 'ggplot2'	0.5.6		
☐	ggthemes	Extra Themes, Scales and Geoms for 'ggplot2'	5.1.0		
☒	ggtree	an R package for visualization of tree and annotation data	3.10.1		
☒	ggtreeExtra	An R Package To Add Geometric Layers On Circular Or Other Layout Tree Of "nntree"	1.12.0		

R Markdown: análisis reproducible



- ❖ Los documentos RMarkdown son ficheros con extensión `.Rmd`.
- ❖ Si acabas utilizando R para hacer análisis estadísticos, los ficheros `.Rmd` te permitirán escribir muy fácilmente informes, tutoriales y transparencias para presentaciones.

Código Rscript

```
chunks.Rmd
1 R Code Chunks
2 =====
3
4 With R Markdown, you can insert R code
5 chunks including plots:
6
7 {r qplot, fig.width=4, fig.height=3,
8 message=FALSE}
9 # quick summary and plot
10 library(ggplot2)
11 summary(cars)
12 qplot(speed, dist, data=cars) +
13   geom_smooth()
```

RStudio: Preview HTML

Preview: ~/chunks.html | Save As | Publish

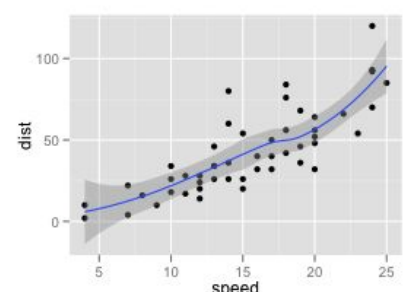
R Code Chunks

With R Markdown, you can insert R code chunks including plots:

```
# quick summary and plot
library(ggplot2)
summary(cars)
```

##	speed	dist
##	Min. : 4.0	Min. : 2
##	1st Qu.:12.0	1st Qu.: 26
##	Median :15.0	Median : 36
##	Mean :15.4	Mean : 43
##	3rd Qu.:19.0	3rd Qu.: 56
##	Max. :25.0	Max. :120

```
qplot(speed, dist, data = cars) + geom_smooth()
```



Reporte

Código
Rscript

chunks.Rmd

Knit HTML

Chunks

```
1 R Code Chunks
2 =====
3
4 With R Markdown, you can insert R code
  chunks including plots:
5
6 ```{r qplot, fig.width=4, fig.height=3,
  message=FALSE}
7 # quick summary and plot
8 library(ggplot2)
9 summary(cars)
10 qplot(speed, dist, data=cars) +
11   geom_smooth()
12 ```
13
```

RStudio: Preview HTML

Preview: ~/chunks.html

Save As

Publish

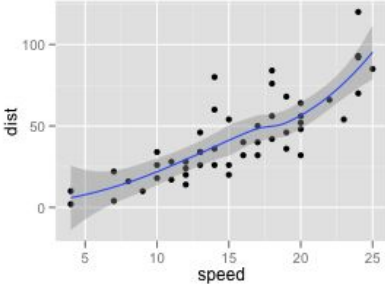
R Code Chunks

With R Markdown, you can insert R code chunks including plots:

```
# quick summary and plot
library(ggplot2)
summary(cars)
```

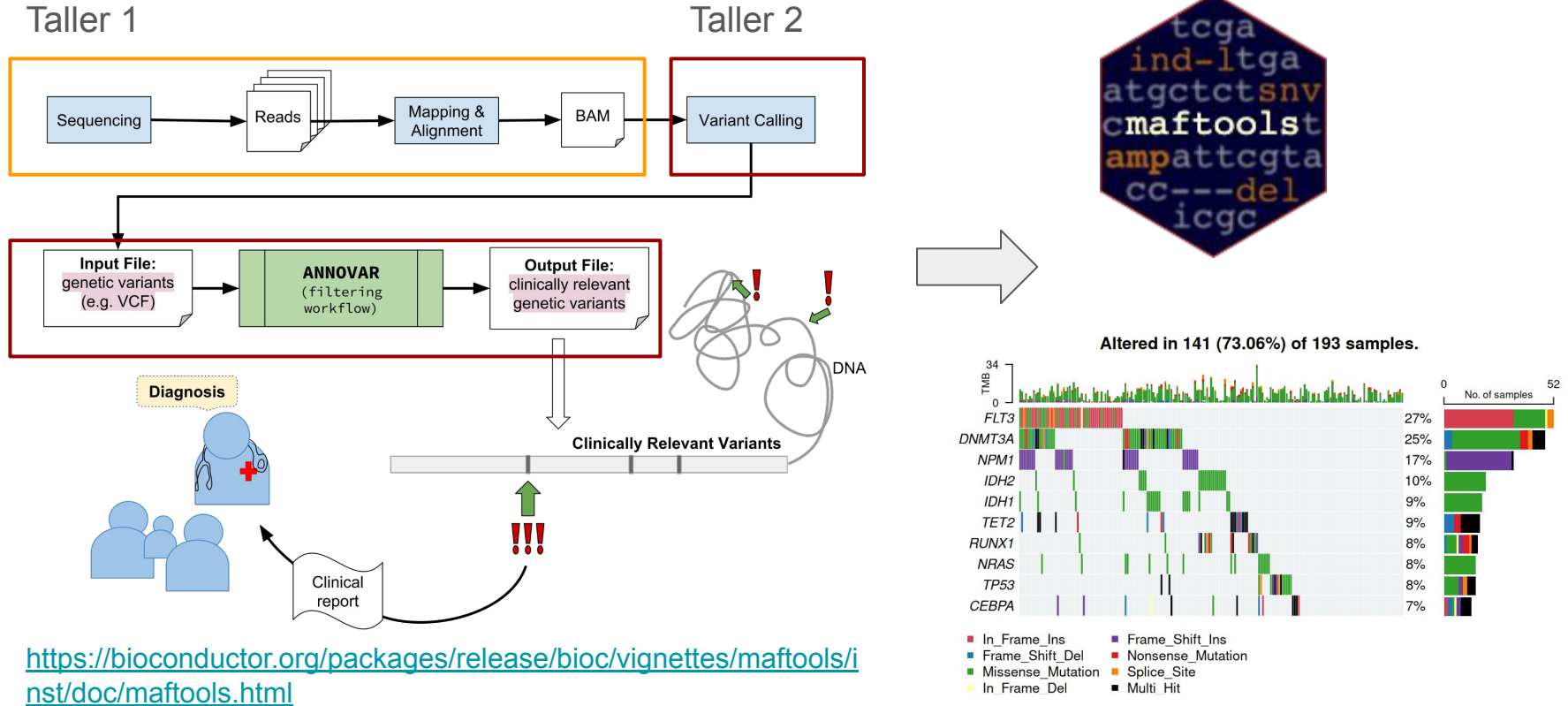
##	speed	dist
##	Min. : 4.0	Min. : 2
##	1st Qu.:12.0	1st Qu.: 26
##	Median :15.0	Median : 36
##	Mean :15.4	Mean : 43
##	3rd Qu.:19.0	3rd Qu.: 56
##	Max. :25.0	Max. :120

```
qplot(speed, dist, data = cars) + geom_smooth()
```



Reporte

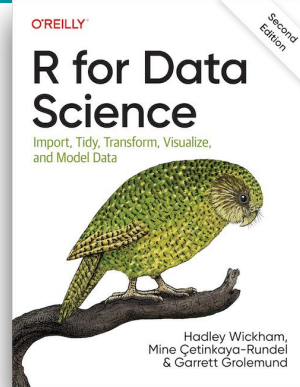
Maftools: Un paquete de R para resumir, analizar y visualizar archivos con mutaciones



<https://bioconductor.org/packages/release/bioc/vignettes/maftools/nst/doc/maftools.html>

Referencias

- ❖ Libro de R de Hadley Wickham
- ❖ R Crash Course for Biologists: <https://colauttilab.github.io/RCrashCourse/>
- ❖ Web: STHDA, datacamp, statology, CRAN
- ❖ Maftools:
<https://bioconductor.org/packages/release/bioc/vignettes/maftools/inst/doc/maftools.html>



R Crash Course
for Biologists
An introduction to R
for bioinformatics and biostatistics

